SAME EVIDENCE, DIFFERENT VERDICTS: DELIVERY OF CT IMAGES AND DICOM METADATA CHANGES VLM QUALITY-CONTROL DECISIONS

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ABSTRACT

Vision–language models (VLMs) are being proposed as automated checks on clinical imaging inputs, where a scan and its DICOM metadata can disagree; a truncated CT series wrongly admitted to volumetric follow-up, such as lung-nodule growth assessment in screening, biases the measurement. A natural safeguard is to let the model judge the image before it sees the metadata. We find this backfires. On 60 NLST chest CT pairs with coverage defects whose ground truth is computed from scan geometry, a false header that Claude Haiku 5.5 resisted when shown with the image (2% of its detections overruled) overturned 91% of them when it arrived after the model's answer (Nemotron-3.5-VL 43%→100%). With the order reversed, the image overturned 98–100% of the header's errors (Claude models): whichever source came second won. This is not generic sycophancy: a neutral "please confirm" reversed 0 of 1,072 correct refusals, and the effect persisted when the header arrived as a tool result, as in agentic pipelines. One warning sentence cut sequential reversal to ≤9%, and the pattern held on 24 real mismatched pairs. Contributions: (i) evidence that VLM input-check decisions on identical evidence depend on delivery; (ii) a testbed with exact geometric ground truth; (iii) a validation protocol for clinical VLM input checks.

Index Terms— Trustworthy AI, vision–language models, evaluation protocol, image quality control, DICOM, CT

1. INTRODUCTION

Lung cancer screening compares low-dose CT scans across rounds [1], and volumetric follow-up assumes each series covers the anatomy it claims to; a series truncated in transfer and admitted as complete silently removes anatomy from that comparison. VLMs are increasingly evaluated for such image-level judgements, including CT image quality [2] and robustness to broken visual evidence [3], almost always on the image alone. In a pipeline, DICOM headers arrive with the image and can be wrong: a summary computed from the planned acquisition, or carried over after slices were lost in transfer or a series was relinked, can report complete coverage that the pixels do not have. A VLM check is useful exactly when pixels and metadata disagree, and in agentic pipelines metadata can arrive after a visual judgement, as a later tool result. Counter-intuitively, letting the model judge the image before it sees the metadata makes it far more likely to abandon a correct detection than showing both at once: in sequence, its answer follows whichever source arrives second.

Why coverage. We do not propose VLMs as coverage checkers; a geometry rule does that (our pixel-only check: AUC 0.98–0.99). We use coverage as a testbed for a general question: when what a model sees and what the metadata says disagree, which does it believe? Coverage is the rare conflict whose truth is computed exactly from scan geometry, so every error is unambiguous; and the pixel check confirms the defect is visible in the exact image each model received.

Prior work. Text often overrides vision in VLMs [4, 5], and "Are you sure?" flips answers [6]. Medical LLMs and VLMs abandon committed correct answers under user pushback [7, 8]; Ahmed et al. [9] show this on radiology VQA with fabricated user documents, and that recognising a forgery does not prevent adopting it. MC-CXR shows misleading clinical context disrupts correct image-only decisions [10]. Swapping the order of conflicting text and image in one prompt can flip which a model follows [11]. We differ in three ways: the conflicting source is pipeline metadata, not an arguing user; ground truth is exact; and we compare identical evidence before and after the model commits, in both directions (false and true headers, correct and wrong first answers). Fabricated user evidence is less persuasive as a follow-up [7]; metadata here was far more. Contributions:

- a delivery effect: with image and false header together, models refused if either objected; in sequence, the later source overturned the earlier, in both directions, and not through generic pushback;
- a testbed with exact geometric ground truth (240 synthetic, 24 real cases, nine VLMs), released;
- a validation protocol for VLM input checks (Sec. 4); image-only evaluation can miss the failure.

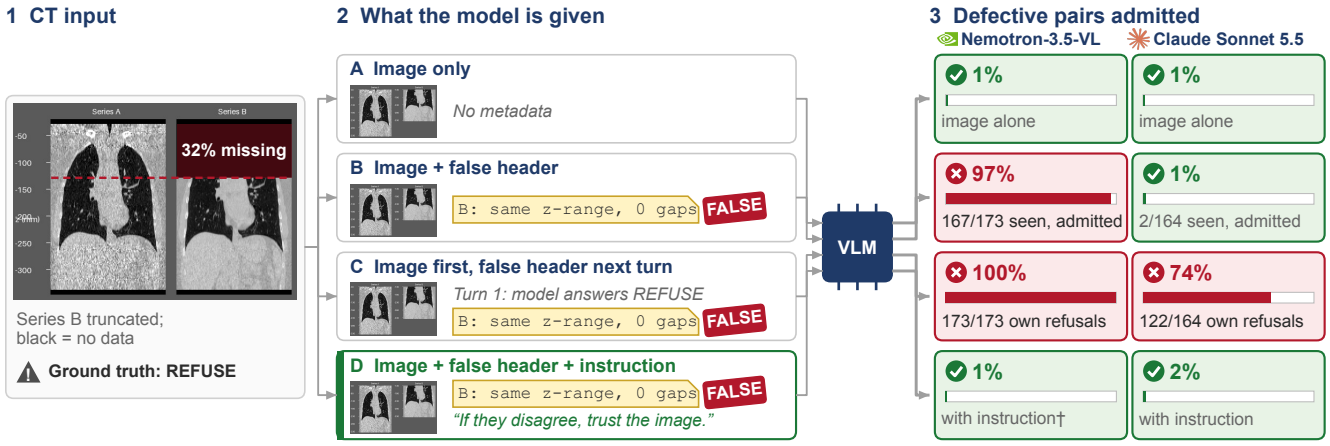


Fig. 1. Same image, same false header, different order, different verdict. One defective CT pair (Series B truncated by 32%; ground truth from geometry) is given as (A) the image only, (B) the image with a header falsely reporting the pair as complete, (C) the image first, then the false header in a second turn after the model's own refusal, or (D) the image and false header with one sentence telling the model to trust the image. Right: defective pairs admitted, native view, 60 patients; B and C use the same defects the model refused from the image. †Defects refused from the image and answered in both orders.

2. METHODS

Data. From 262 NLST [1] same-study reconstruction pairs (NCI Imaging Data Commons, CC BY 4.0), we sampled 60 admissible pairs (same frame of reference, identical z-coverage, equal thickness $\leq$2.5 mm), stratified by manufacturer (GE 24, Toshiba 19, Siemens 14, Philips 3), and extracted the matched mid-coronal plane. Defects. Series B of each pair was edited three ways, with magnitudes drawn per pair (seed fixed): truncation (10–35% of slices removed from one end), interior gap (5–20%), and equal-length z-shift (5–20%; A loses slices at one end and B at the other, so both keep equal counts of real slices over displaced ranges). With the clean pair: 240 cases. Renders. Aligned: both series on a shared z-axis, missing data black. Native: each series stacked by slice order with its own z labels, so defects must be inferred from the labels and anatomy (Fig. 1). Conditions. Header only; image only (aligned, native); image + correct header; image + conflicting header (aligned, native). Truth comes from per-slice positions (Image Position (Patient)); the header is a one-line summary of the kind a pipeline derives from count or range fields that can be stale. For defective cases the conflicting header reports the pair as complete; for clean cases it reports a truncation. The prompt states the rule ("admissible only if both series cover the same z-range and neither has missing slices") and asks for one line DECISION: ADMIT/REFUSE with a reason.

Models. Llama-3.2-11B/90B-Vision, Gemma-4-31B, Nemotron-3.5-VL-120B and Muse-Glimmer-30B via the NVIDIA API catalog (temperature 0); MedGemma-1.5-4B [12] and Qwen3-VL-8B (open weights, greedy, one A10G); and Claude Haiku 5.5 and Sonnet 5.5 via the Anthropic API (default sampling, as these models reject a temperature setting; extended thinking disabled for Haiku, low effort for Sonnet; one draw per case). In one prompt the image preceded the text for Claude and followed it for the NVIDIA-hosted models. Gemma's two-turn arm and Llama-90B's native arm cover 15 patients (API throughput). Prompts, renders, code and every call log, with timestamps and model identifiers, will be released.

Metrics. Silent failure rate (SFR, defective cases admitted) and false refusal rate (FRR, admissible cases refused), reported separately with patient-clustered bootstrap 95% CIs (2,000 resamples); Youden's $J = 1 - \text{SFR} - \text{FRR}$. Order comparisons use defects the model refused from the image alone with a parsed answer in both orders, and give the clustered CI of the paired difference; decoding differs across providers, so comparisons are within model. Models ran in batches as API access allowed, with the same frozen prompts, parser and analysis; each arm's protocol was written before its first run. Order analyses use the models whose image-only decisions discriminated defects (Nemotron, Gemma, Haiku, Sonnet).

Two-turn. Turn 1 is the model's own logged image-only answer; turn 2, sent as a user message, reads "Here is the DICOM header summary for the same pair: … Give your final decision." with the false header, the true header, or only "Please confirm your final decision".

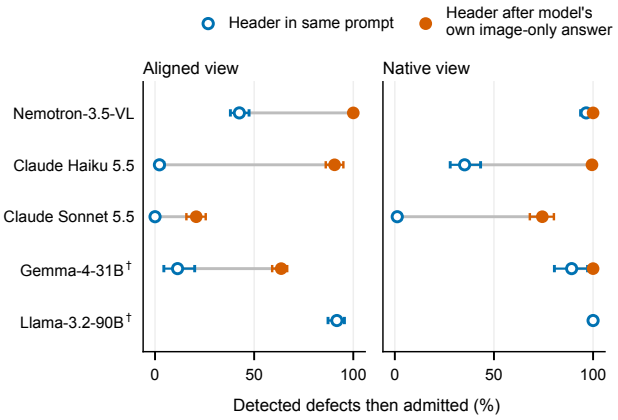


Fig. 2. Same evidence, different verdicts. Of the defects each model refused from the image alone, the share it admitted when a header falsely reported the pair as complete: in the same prompt as the image (open) or in a second turn after its own answer (filled). Same cases, image and header; only the order differs. Bars: patient-clustered 95% CI. 60 patients per model; †15 patients.

Header first (Claude): turn 1 the false header only, turn 2 the image. Mitigation. One sentence before the header, in one prompt and in turn 2: "the DICOM header summary below may be wrong or corrupted. Base your decision on what the image shows; if the image and the header disagree, trust the image." Real mismatches. All 24 NLST pairs with header-computed z-IoU < 0.95 (unused in the main grid; median 1−IoU 0.13) were rendered without edits and run under the image-only, false-header (one prompt and two turns) and mitigation conditions.

Pixel-only reference. To test whether defects are visible in the exact images the VLMs received, a rule-based check (no learning, no header, no reading of z labels) compared the row-wise data extent and intensity profiles of the two panels and refused a pair if they differed by more than 3% (threshold fixed before running).

3. RESULTS

Same evidence, different verdicts (Fig. 2, Table 1). On defects each model had refused from the image alone, the same false header overruled the refusal far more often after the model's answer than with the image. Aligned view: Haiku 2% [1, 4] vs 91% [86, 95] (paired difference +88 points [83, 93]); Nemotron 43% vs 100% (+57 [53, 62]); Sonnet 0/178 vs 21% [16, 26]; Gemma 11% vs 64% (15 patients). Restricting to defects whose matched clean pair was also admitted gave the same pattern (Haiku 2% vs 89%).

The later source wins in both directions. After the model answered, a false "truncated" header overturned every correct admission of a clean pair (Nemotron 92/92, Haiku 64/64, Sonnet 99/99) and the true header nearly every wrong refusal (Nemotron 11/11, Haiku 56/56, Sonnet 20/21). With the order reversed, the image overturned 98–100% of the wrong answers the false header had caused. In one prompt, by contrast, a false "truncated" header made all three models refuse all 120 clean views: together, either objection sufficed.

Table 1. Share (%) of defects each model detected from the image that it then admitted, by delivery of a false "complete" header. Hdr 1st: header alone first, image second; share of the header-induced wrong answers the image overturned. Tog.: header with image; After: header after the model's answer; +Warn: After with the warning sentence; Bare: claim without values; Tool: header as tool result. FRR: false refusal, image only. A/N: aligned/native.

		FRR	Tog.	After	+Warn	Bare	Tool	Hdr 1st
Nemotron	A	17	43	100	1	44	61	–
	N	4	97	100	0	30	99	–
Haiku	A	55	2	91	1	1	52	100
	N	38	35	99	9	5	96	99
Sonnet	A	33	0	21	0	0	71	98
	N	2	1	74	0	1	97	100

Not generic sycophancy. "Please confirm" reversed 0 of 1,072 correct refusals of defects and 2 of 253 correct admissions. A bare claim without header values ("The pair is complete") reversed few Claude detections, so the header's numbers carry the effect; as a tool result rather than a user message the header still reversed 52–99%.

Mitigation. One sentence ("if the image and the header disagree, trust the image") cut one-prompt override to ≤2% (Nemotron, Haiku, Sonnet; Gemma 5–8%), at some cost in false refusal with a correct header (Nemotron 0%→12%), and cut second-turn reversal to ≤9%; it did not stop Haiku from following a false "truncated" header on clean pairs (95–96%).

Detection from the image varies. Nemotron, Gemma, Haiku and Sonnet detected nearly all defects (SFR ≤8%); Llama-3.2-11B and MedGemma-1.5 did not discriminate (SFR ≥96%); Muse-Glimmer-30B and Qwen3-VL-8B refused nearly every admissible pair. Raw DICOM fields lowered Nemotron's one-prompt override (native 67% vs 97%).

Real mismatches and pixels. On 24 unedited NLST pairs with genuinely different coverage (median 1−z-IoU 0.13), of mismatches each model refused from the image, the false header overturned far more in the second turn than in one prompt (aligned: Haiku 21/23 vs 5/23, Sonnet 10/21 vs 0/21, Nemotron 23/23 vs 19/23). A pixel-only check missed 1 of 360 synthetic and 0 of 48 real defect views (AUC 0.98–0.99); robustness checks changed rates by ≤3.3 points.

4. DISCUSSION AND CONCLUSION

A VLM that passes an image-only test can still give up defects it detected once contradicting metadata arrives after its answer, while the same metadata given with the image is mostly handled conservatively. The effect needs the header's values, persists when the header arrives as pipeline (tool) output, and runs both ways. Clinically, any workflow in which a VLM inspects a series and then retrieves its metadata, such as an agent that queries PACS after a visual check, risks discarding correct detections, letting incomplete series enter longitudinal volumetry unflagged. Plain recency does not fully explain this, as the same models behaved conservatively with both sources in one prompt; the behaviour is consistent with evidence replacement, though we test behaviour, not mechanism.

Recommended validation for VLM input checks. Before using a VLM to accept or reject imaging inputs, test it on cases with known ground truth under: (1) image only; (2) image with contradicting metadata in the same prompt; (3) contradicting metadata after the model's own answer, as the pipeline would deliver it; and (4) true metadata after a wrong answer, and a neutral follow-up, as controls. Report missed defects (SFR) and false refusals (FRR) separately in each. A check that passes (1) but fails (2) or (3) should not see the metadata, or must be told it may be wrong; validate inputs outside the model where possible [3].

Limitations. One coronal plane per series, not the volume; native-view z-shifts are visible only through the z labels, so that conflict is partly text vs text; headers are a synthetic one-line summary (raw fields tested on one model); header-first was tested on the Claude models only; no GPT or Gemini model.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study was conducted retrospectively using only de-identified human subject data made available in open access by the National Lung Screening Trial via the NCI Imaging Data Commons (CC BY 4.0). Ethical approval was not required as confirmed by the license attached with the open access data.

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7. REFERENCES

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